Investigating Physical Properties of Granular Materials using Network Science

NS Mini Project

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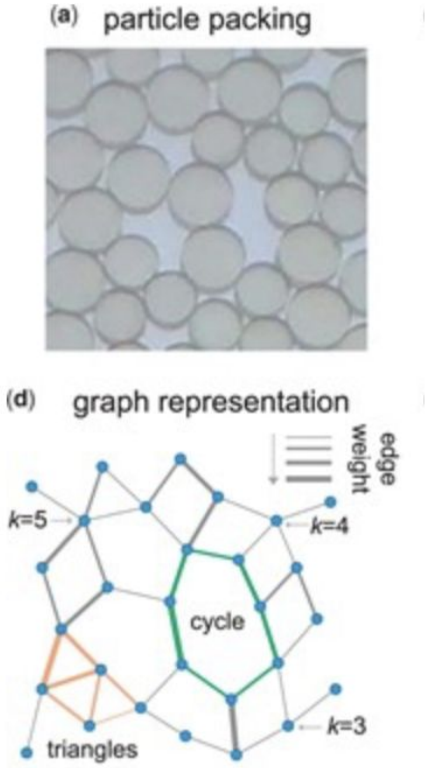
Introduction and Background

- Granular material are collections of discrete solid particles (e.g., sand, grains) that exhibit both fluid- and solid-like behavior depending on external forces.
- Granular materials are ubiquitous in nature and industry, yet their collective behavior for example jamming,flow and failure remains complex and non-linear.
- Network science techniques can be used to model and analyze complex systems such as granular materials by representing them in the form of a graph where each particle represents a node and each edge represents the interactions between physically adjacent particles.
- It enables a quantitative and scalable analysis of the structure of the material and Links microscale contact patterns to macroscale properties like rigidity, strength, and failure.

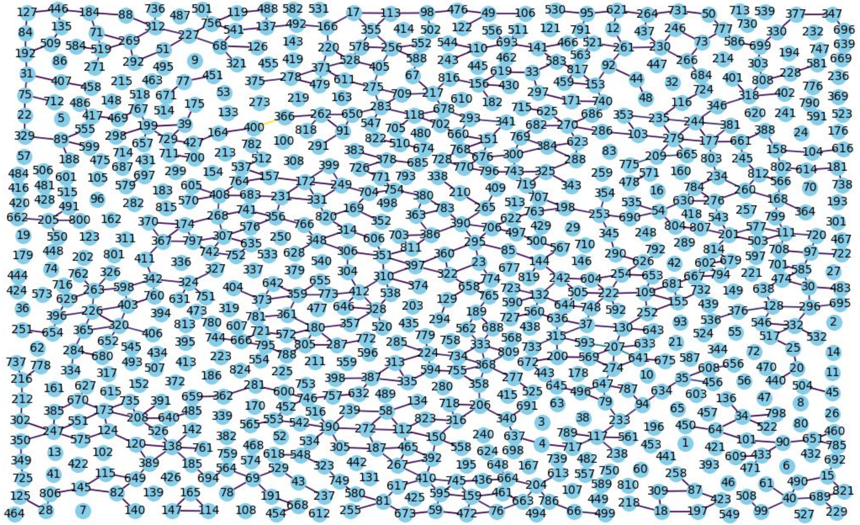


Correlation with Physical Properties

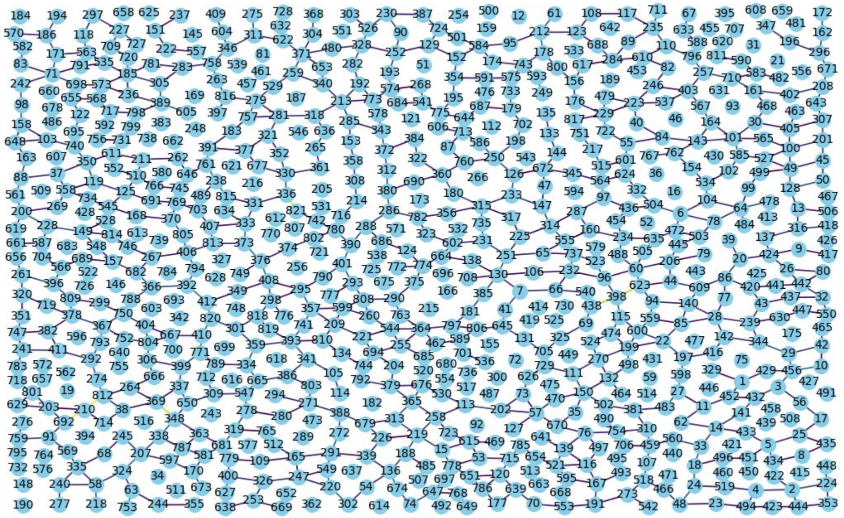
- Various parameters used in network analysis average degree, clustering, connected components, modularity, betweenness, etc. Each correlates with physical aspects such as force propagation, stability, or deformation paths.
- Network based analysis of granular materials can help in selecting the best materials based on task specific requirements and aid in designing novel materials with desired properties.
- In this presentation we compare two different granular materials and showcase how each network property influences the physical nature of the material.
- The correlation of each network parameter with physical properties is defined in the following research paper : https://academic.oup.com/comnet/article/6/4/485/4959635

Network Visualization

Material-1



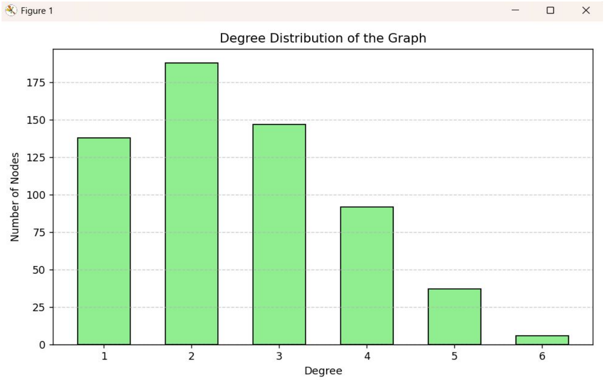
Material-2

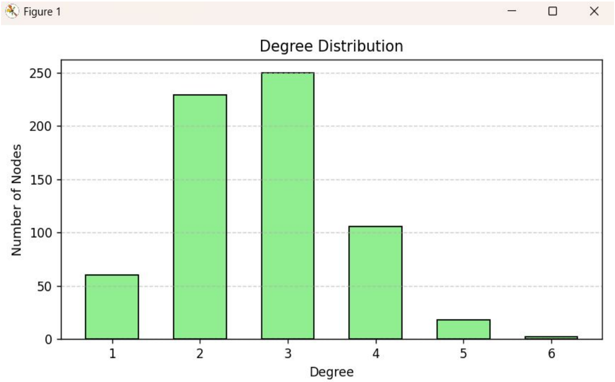


Obtaining Networks using DEM

- We initially ran DEM (Discrete element method) simulation which is a numerical technique used to simulate the interaction between a large number of particles governed by Newton’s laws of motion using an open source software (LIGGGHTS)
- Choose two hypothetical granular materials with varying particle sizes, shapes, density, and material properties such as Young's modulus and coefficient of restitution.
- Took a screenshot of the materials in time and constructed respective 2D cross sectional networks with the help of the dump files generated by the simulation tool.

Degree Distribution and Average Degree





Material-1 : Average Degree = 2.5395

Material-2 : Average Degree = 2.6977

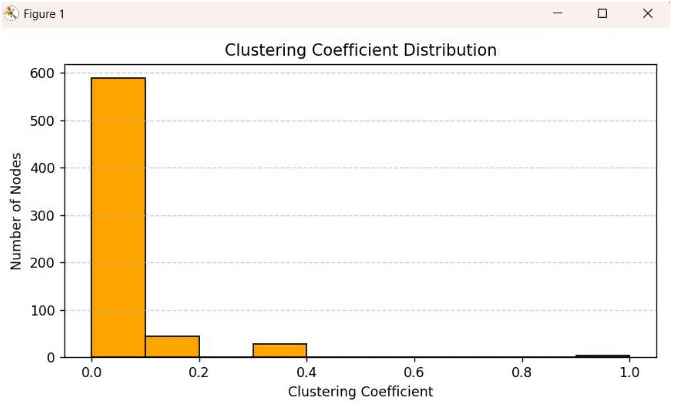
- A greater average degree in the case of a granular network corresponds to higher packing density and greater mechanical strength which material-2 has more of in comparison to material-1.
- The higher the degree of a node the more particles it is in contact with and due to this it has more avenues to distribute any force more evenly on its neighboring particles in physical contact.
- Upon inspection of the degree distribution, we can observe that the number of nodes with degrees 2,3 and 4 for material-2 is significantly higher than than material-1 leading to the conclusion that material-2 has more such nodes which can distribute forces efficiently making it more stable.

Clustering Coefficient Distribution

Material-1

Average Clustering Coefficient: 0.0291

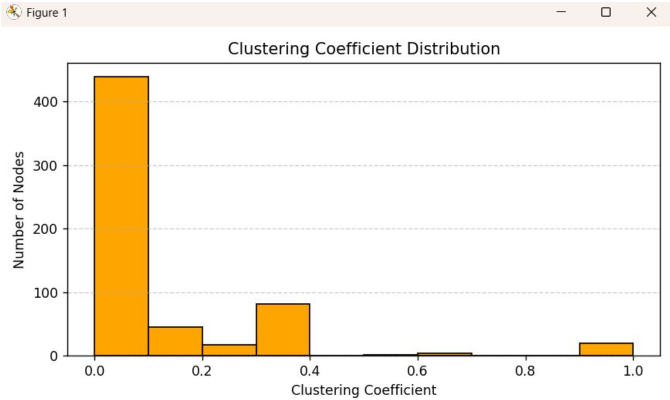
Global Clustering Coefficient (Transitivity): 0.0444



Material-2

Average Clustering Coefficient: 0.1003

Global Clustering Coefficient (Transitivity): 0.1371



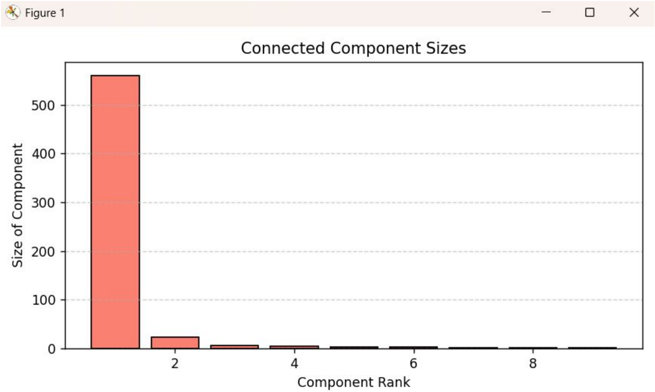
- A higher average clustering coefficient indicates more mechanical stability and resistance to failure upon application of external force or stress which is more for material-2 as compared to material-1.
- A high global clustering coefficient indicates the network wise prevalence of 3 node (triangular) cycles which confer stability to the material because of their inherent physical organization, this indicates that material-2 has a higher number of 3 node cycles as compared to material-1 making it more robust to external stress.

Connected Components

Material-1

Number of Connected Components: 9

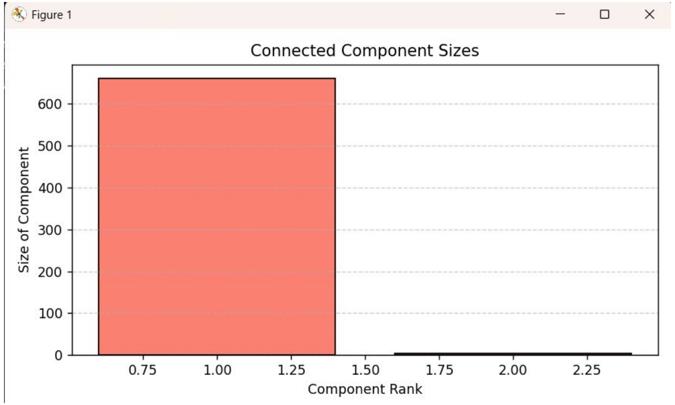
Largest Component Size: 560



Material-2

Number of Connected Components: 2

Largest Component Size: 661



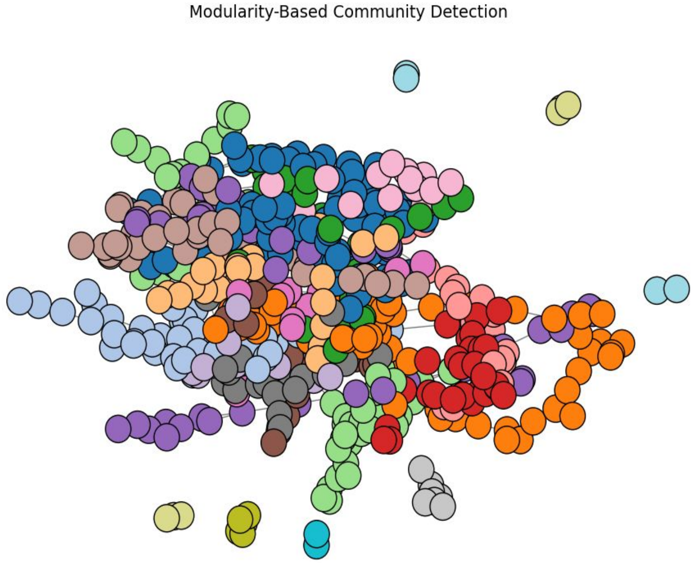
- If the number of connected component are small and the size of the largest component is sufficiently large then :
  - A large, connected network can transmit forces globally while many small groups can't leading to better mechanical stability
  - Large connected structures share and distribute stress across the system leading to enhanced stress propagation.
  - Small components fail independently or cannot resist external forces
  - High elastic modulus or bulk strength
- In this regard material-2 performs better on the above benchmarks by comparison of these network parameters.

Modularity-Based Community Detection

Material-1

Modularity of the graph : 0.9007

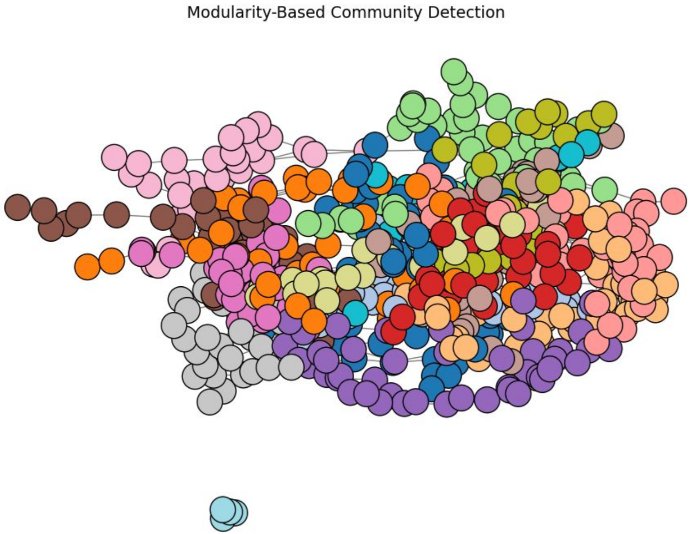
Number of communities found: 28



Material-2

Modularity of the graph : 0.8438

Number of communities found: 17



Inferences

- Structure 1:
  - Modularity = 0.9007 with 28 communities
  - Structure 1 splits into more, tighter groups that hardly connect to each other.
- Structure 2:
  - Modularity = 0.8438 with 17 communities
  - Structure 2 makes fewer, bigger groups with more “cross-talk” between them.

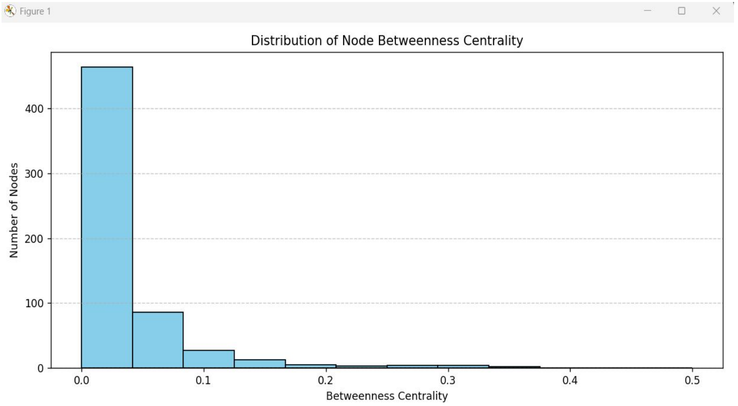
What it means:

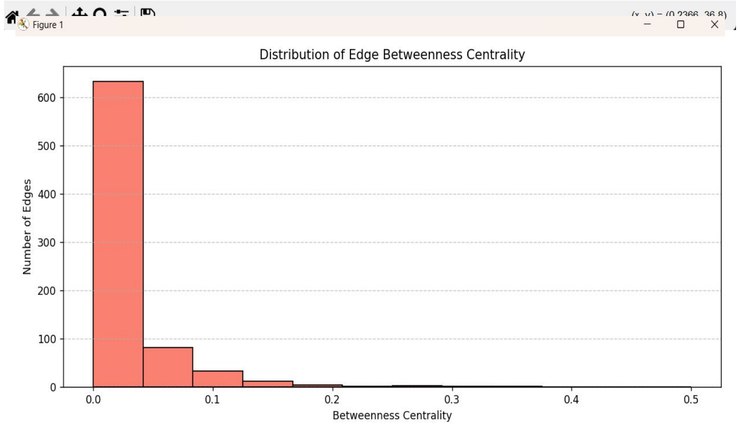
- In Structure 1, something (like information, heat, failures…) stays inside one group longer before jumping out.

- In Structure 2, it’s easier to go from one group to another, so things spread faster across the whole network.

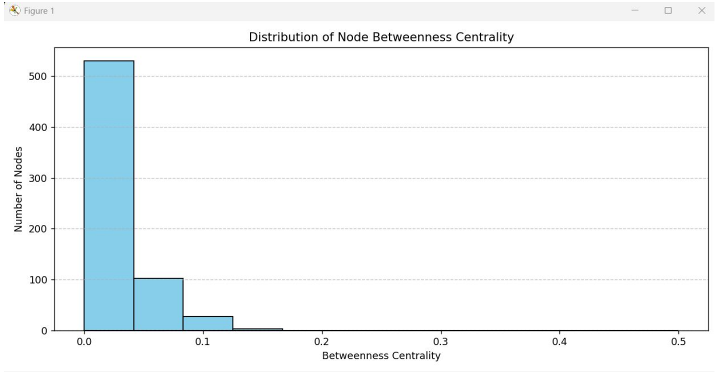
Betweenness Centrality Of Nodes and Edges

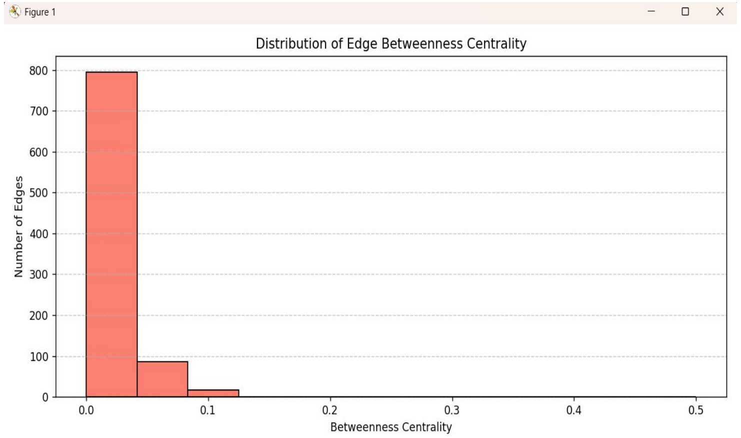
Material-1





Material-2





Inferences

Node Betweenness

- Histograms for both show a heavy-tailed distribution:
  - Most nodes have near-zero betweenness (they sit on few shortest-paths).
  - A few nodes have very high betweenness (critical bridges).
- Structure 2 has a slightly larger mass right at the zero-bin and a marginally steeper drop into the tail.

Edge Betweenness

- Similarly heavy-tailed, with Structure 2 again showing:
  - More edges with essentially zero betweenness,
  - A tiny handful of edges carrying the bulk of shortest-path traffic.

What it means:

Structure 1 has a few nodes/edges with relatively higher importance—these can serve as backup routes if a main bridge fails.

Structure 2 piles almost everything onto its top few carriers—so if one of those breaks, there's no easy detour.

Additional parameters

Structure-1	Structure-2
Global Efficiency: 0.0608	Global Efficiency: 0.0795
Mean Shortest Path Length: 21.8248	Mean Shortest Path Length: 17.8598
3-cycle to 4-cycle ratio: 0.7714	3-cycle to 4-cycle ratio: 2.0270

Global Efficiency: It is a measure of how efficiently information or interaction can be exchanged over a network. It reflects the average inverse shortest path length between all pairs of nodes. Structure-2 is more globally efficient, indicating faster communication or transport between distant nodes. This may reflect lower spatial or energetic cost for efficient transfer of heat or sound waves.

$$\mathrm{E}_{glob}(G) = \frac{1}{N(N-1)} \sum_{i \neq j i, j \in G} \frac{1}{d_{ij}}$$

A shorter **average path** in Structure-2 indicates better connectivity and less resistance for transport or signal propagation, consistent with the higher global efficiency.

A higher **3-cycle to 4-cycle ratio** in Structure-2 suggests a denser local connectivity (triangles), implying greater clustering and potentially better robustness or spatial compactness. Three node cycles are structurally stable while four node cycles are weak and floppy.

Other Way of Representing Granular Networks

Force-Weighted Network Representation

Definition: A network where each node represents a particle, and edges represent physical contacts weighted by the magnitude of inter-particle forces.

Purpose:

Captures mechanical interactions beyond simple connectivity.

Highlights the role of force chains — sequences of particles bearing higher-than-average force.

Analysis Directions:

Cluster analysis on high-weight edges to identify strong force chains.

Detect force bottlenecks — regions where large forces are funneled through few particles.

Identify weak zones or failure-prone areas by locating edges with low force weights or high betweenness.

Applications:

Understanding jamming transitions, shear failure, and stress localization.

Informing design of stable granular packings in materials and civil engineering.

Thank You